

Introduction

The primary objective of this project is to predict whether a student will graduate, remain enrolled, or drop out, based on academic, demographic, and socioeconomic factors. This assignment focuses on fairness-aware machine learning. This means, in addition to high accuracy, the system should ensure that predictions are not biased against specific groups, such as gender or nationality.

Dataset Summary

We used the dataset "Predict students' dropout and academic success" from the UCI Repository.

Number of records: 4,424 students.

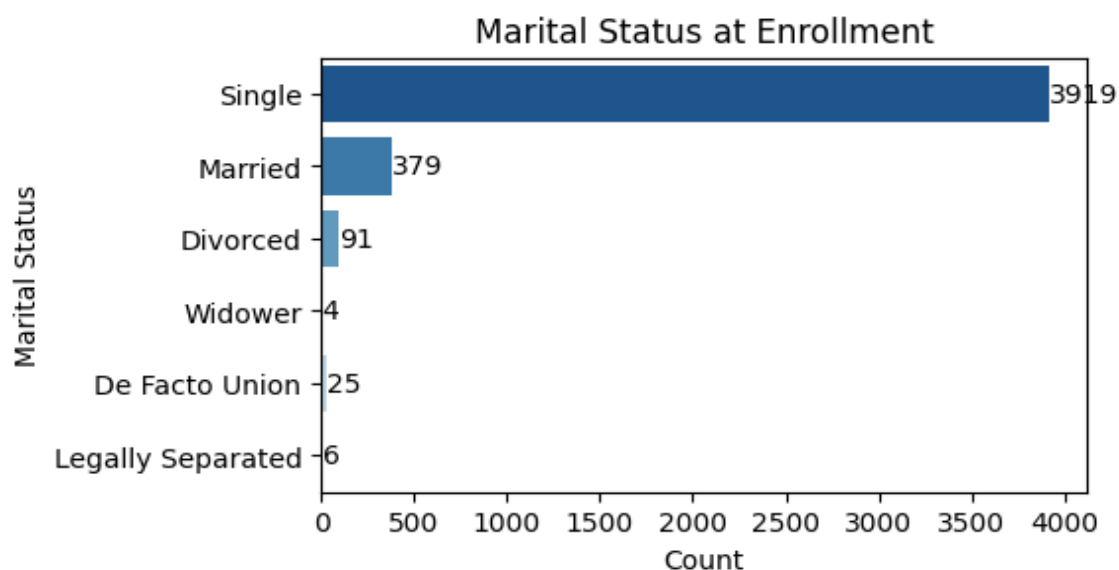
Number of features: 36 input features and 1 target label.

Target variable:

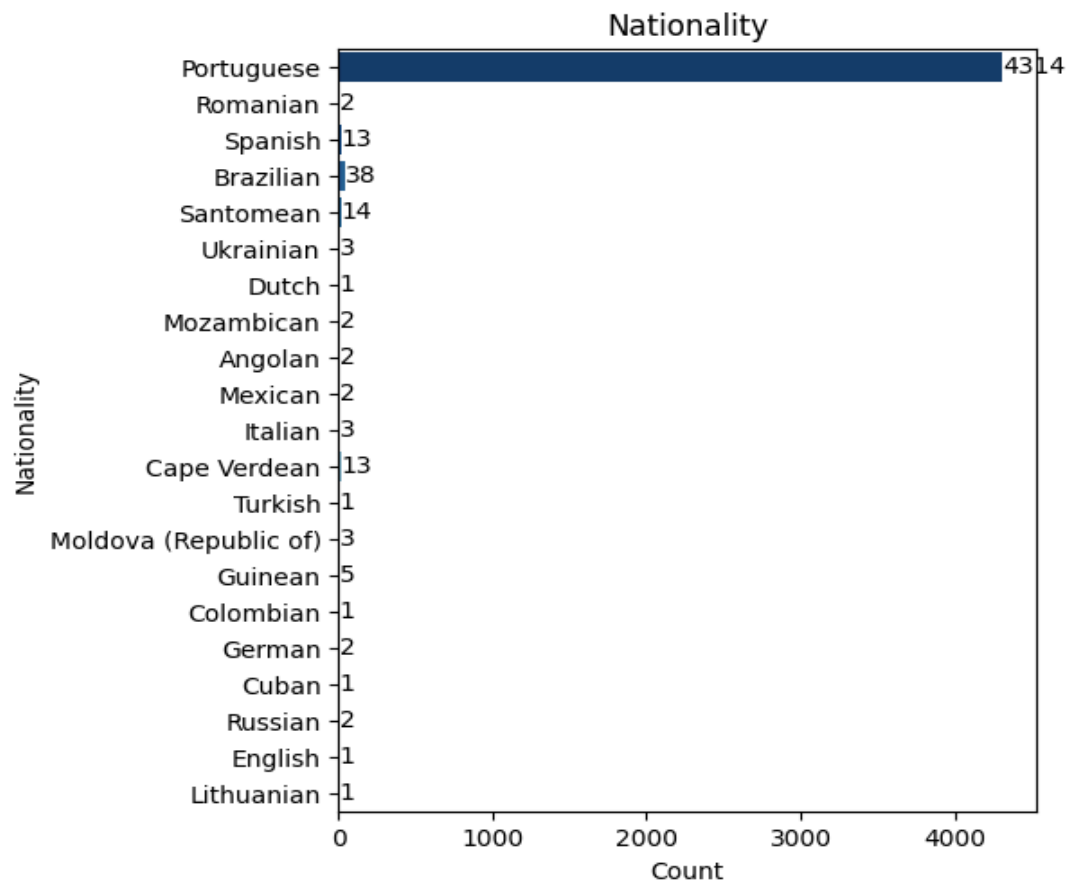
- Dropout (student left the course),
- Enrolled (student is continuing),
- Graduate (student completed the program).

Feature Groups

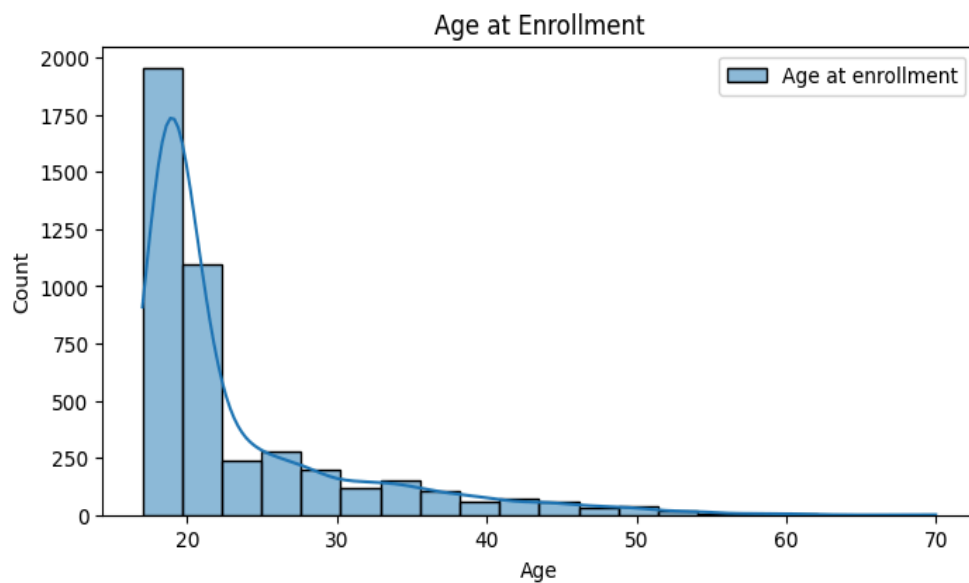
- Demographic: Gender, age, marital status, nationality.



The majority of students are single, followed by married, and then divorced. Only a small proportion fall under categories such as de facto union, legally separated, or widowed.

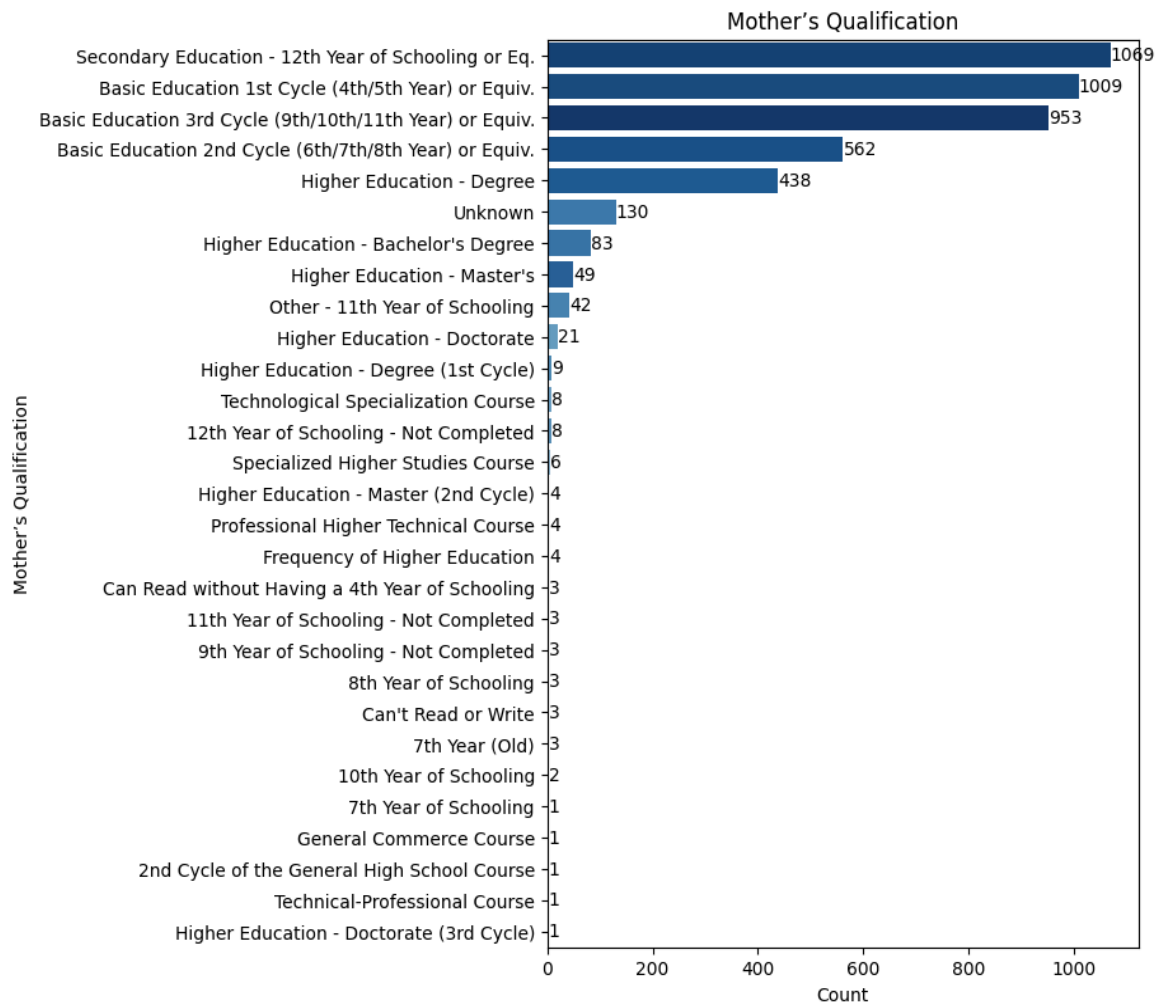


Most of the students are from Portugal, followed by Brazil

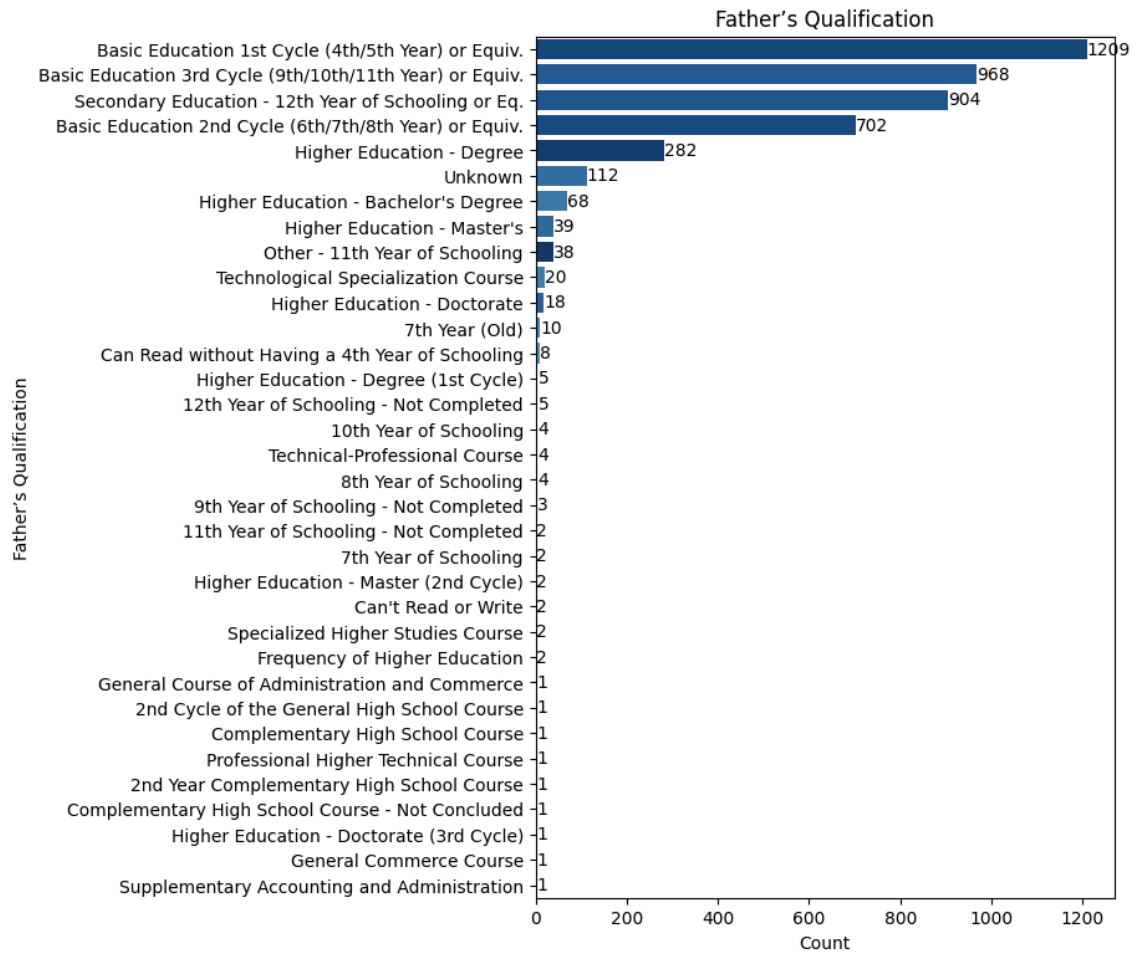


Most students are young, while very few above the age of 30 pursue studies.

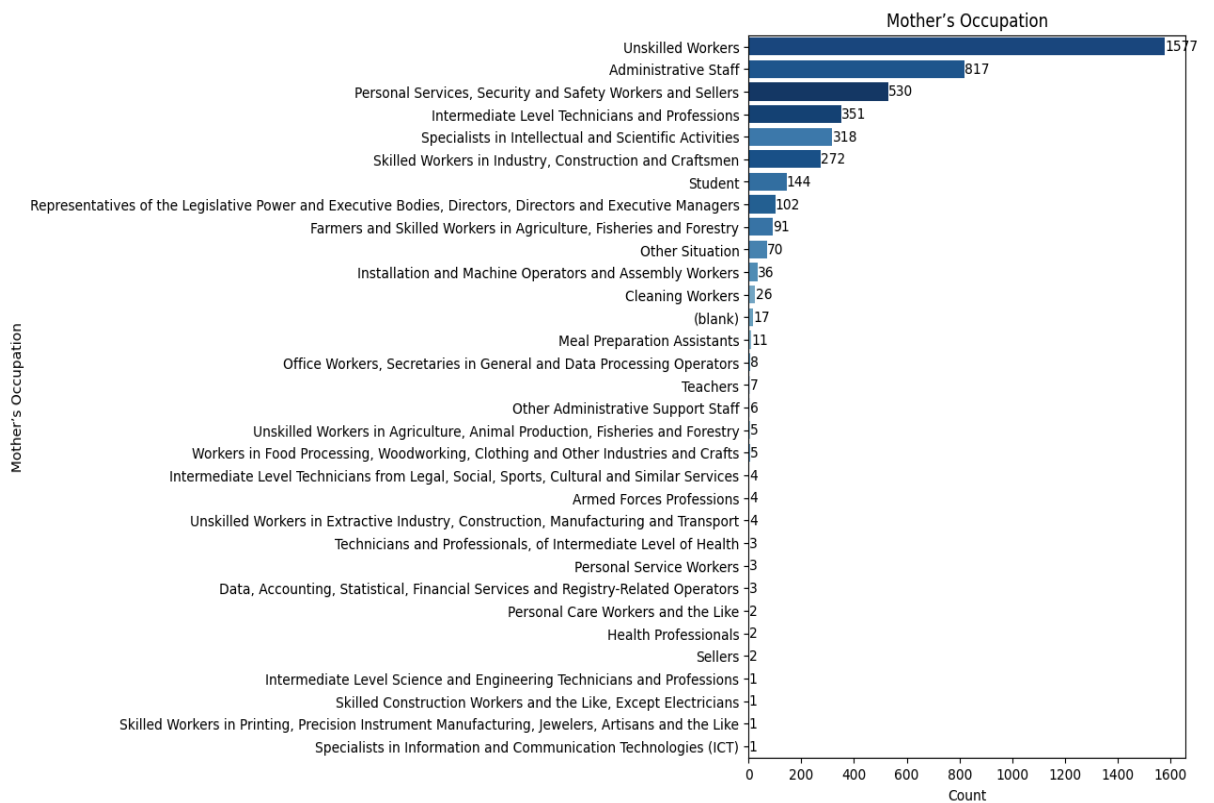
- Socio-economic: Parents' education, occupation, scholarship holder, debtor status, and tuition fee up to date.



Most of the mothers' qualifications are Basic Education, followed by Secondary Education. Very few have a Degree.

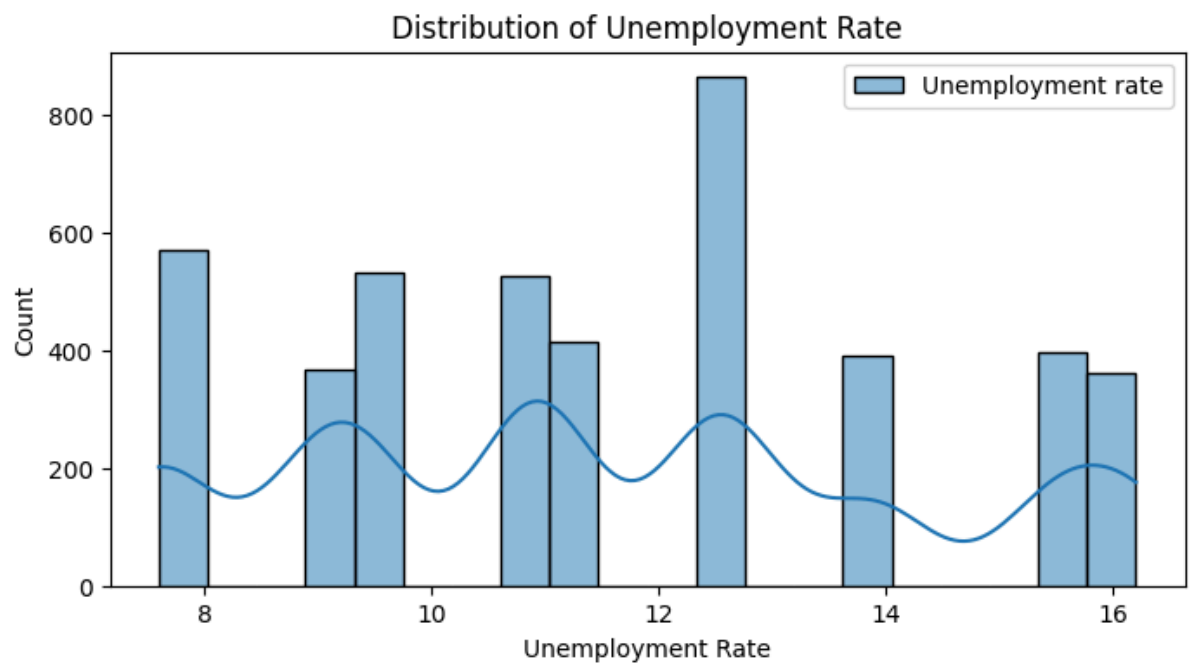


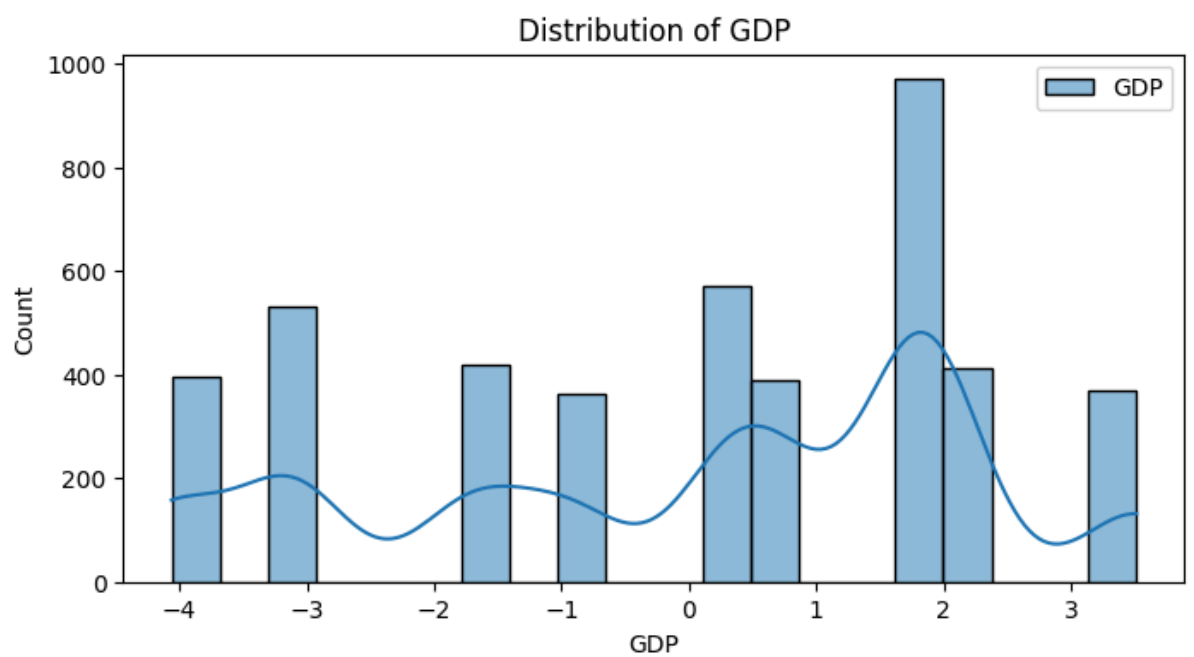
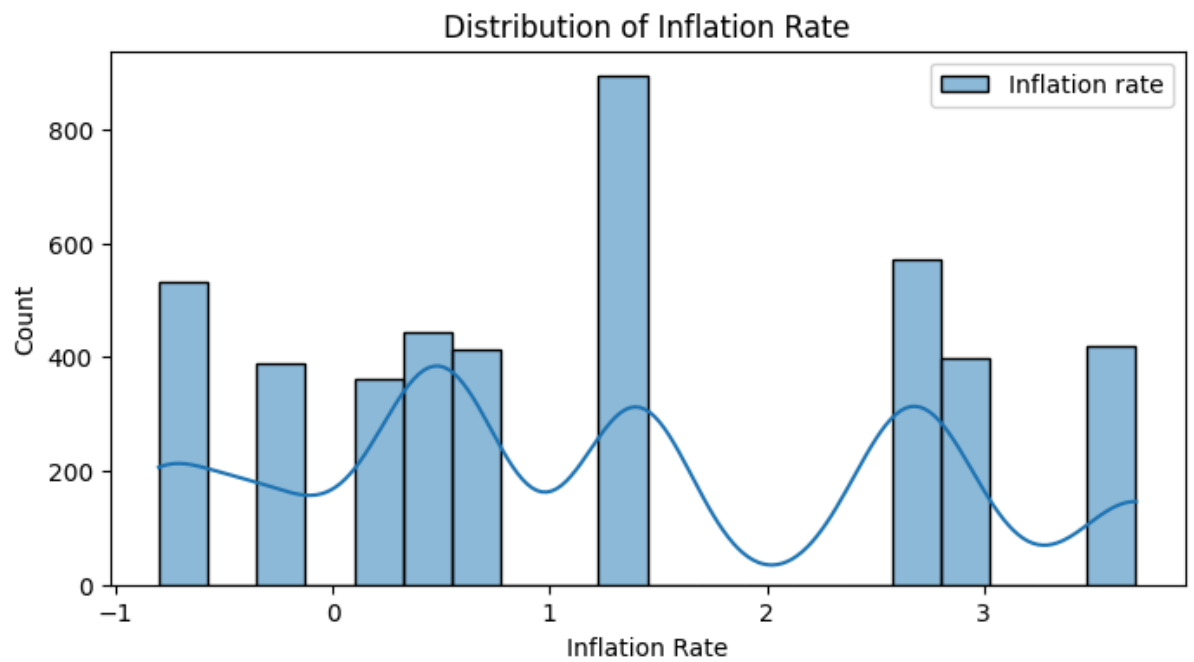
Most of the Father's qualifications are Basic Education, followed by Secondary Education. Very few have a Degree.



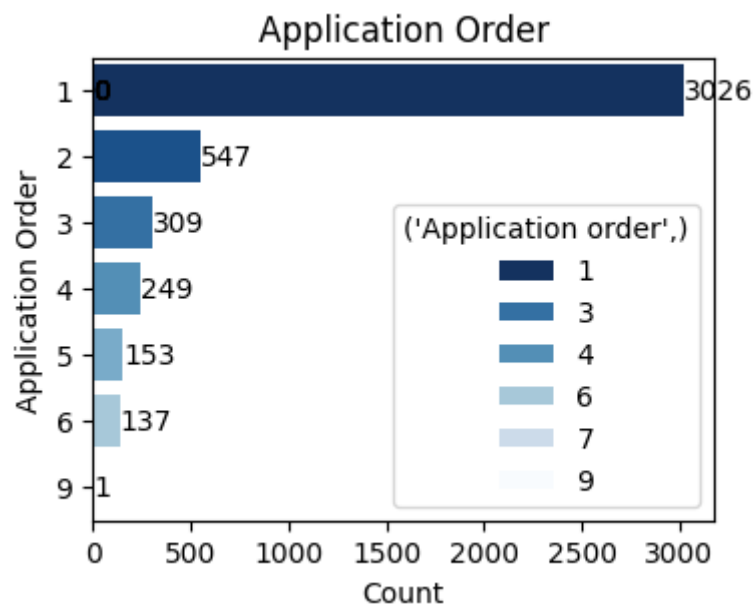
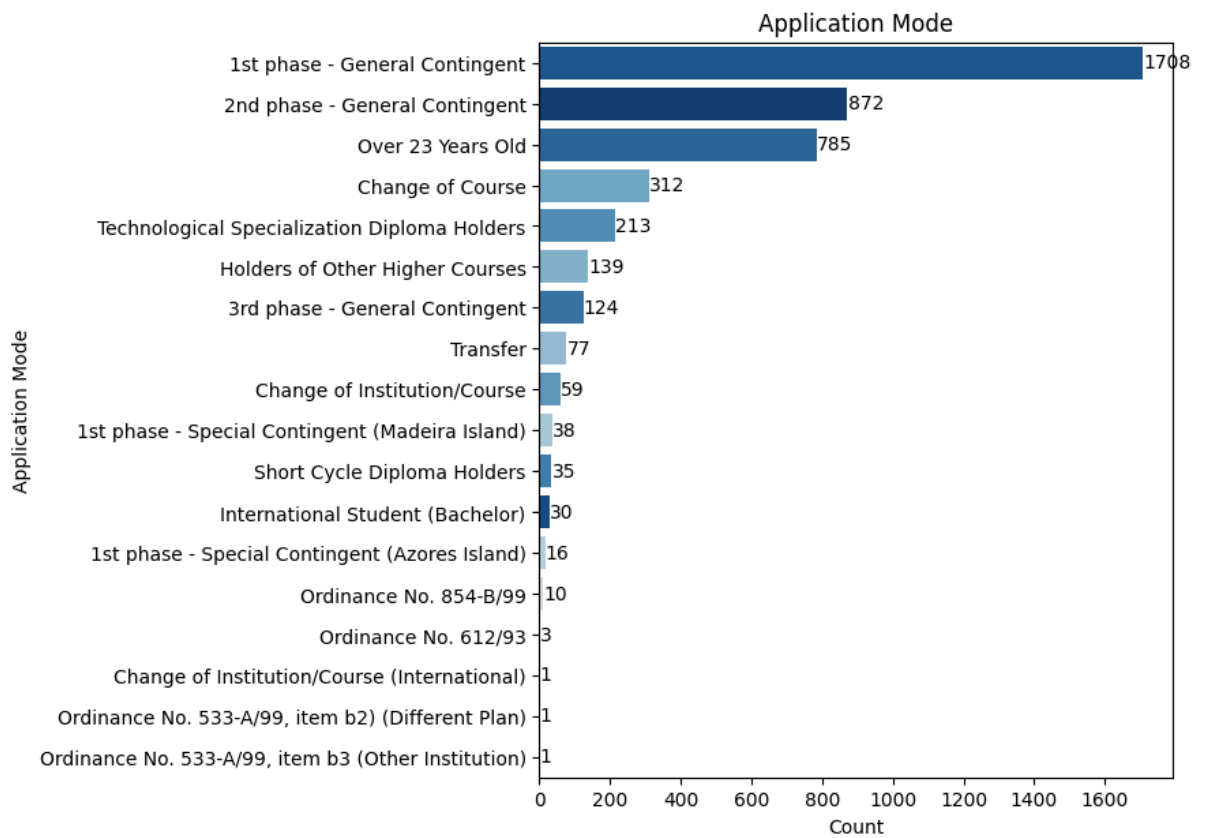
Mostly Mothers are “Unskilled workers”, followed by “Administrative Staff” and “Personal Services, Security and Safety Workers and Sellers”

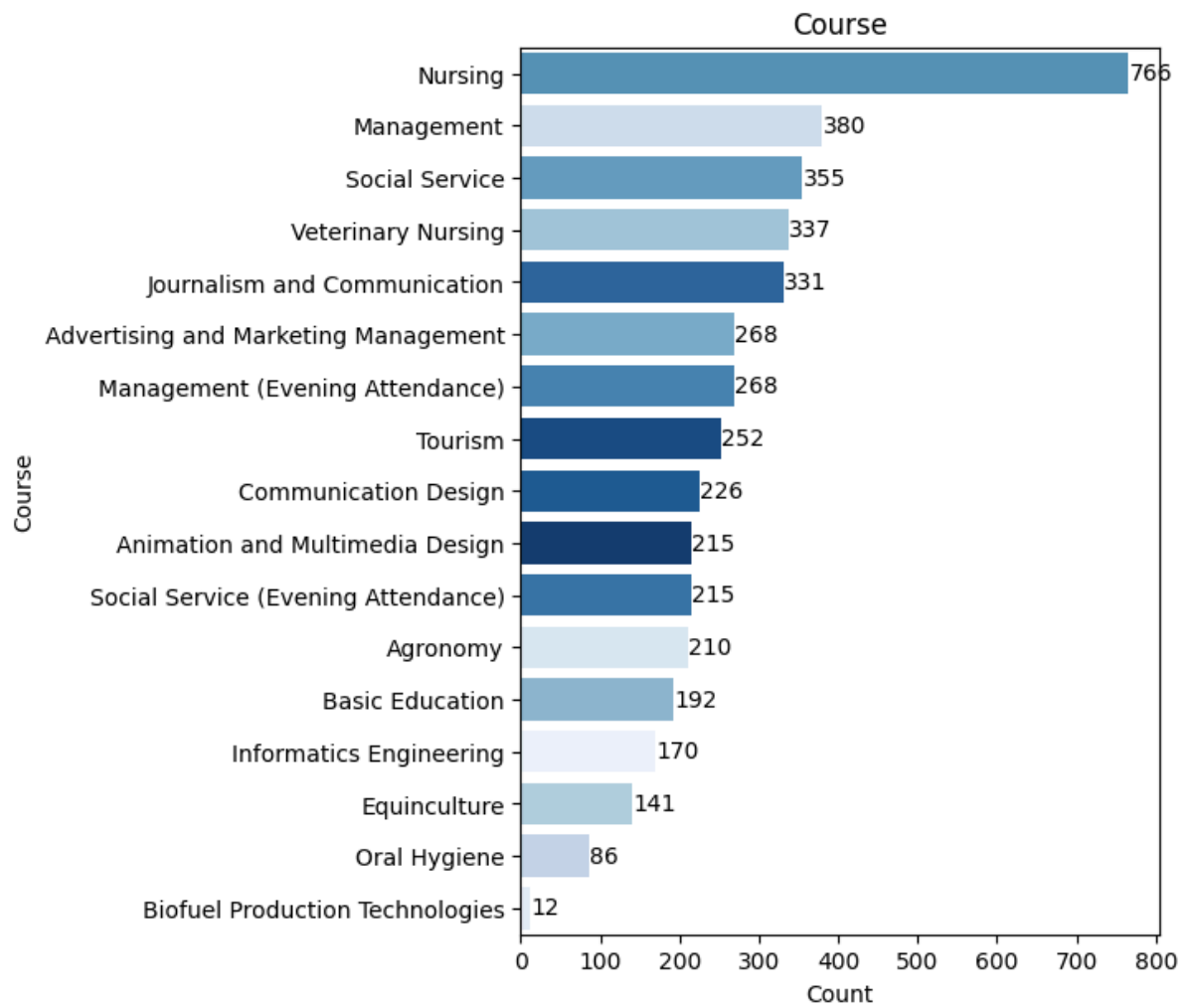
- Economic: Country-level stats like GDP, unemployment rate, and inflation.

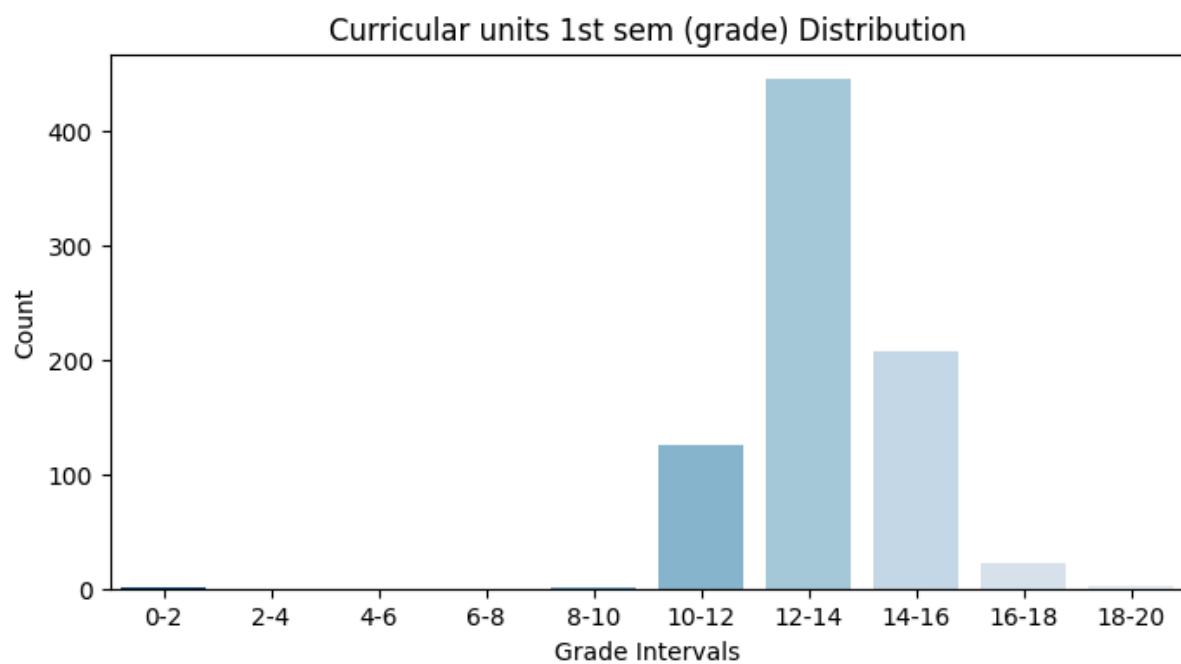
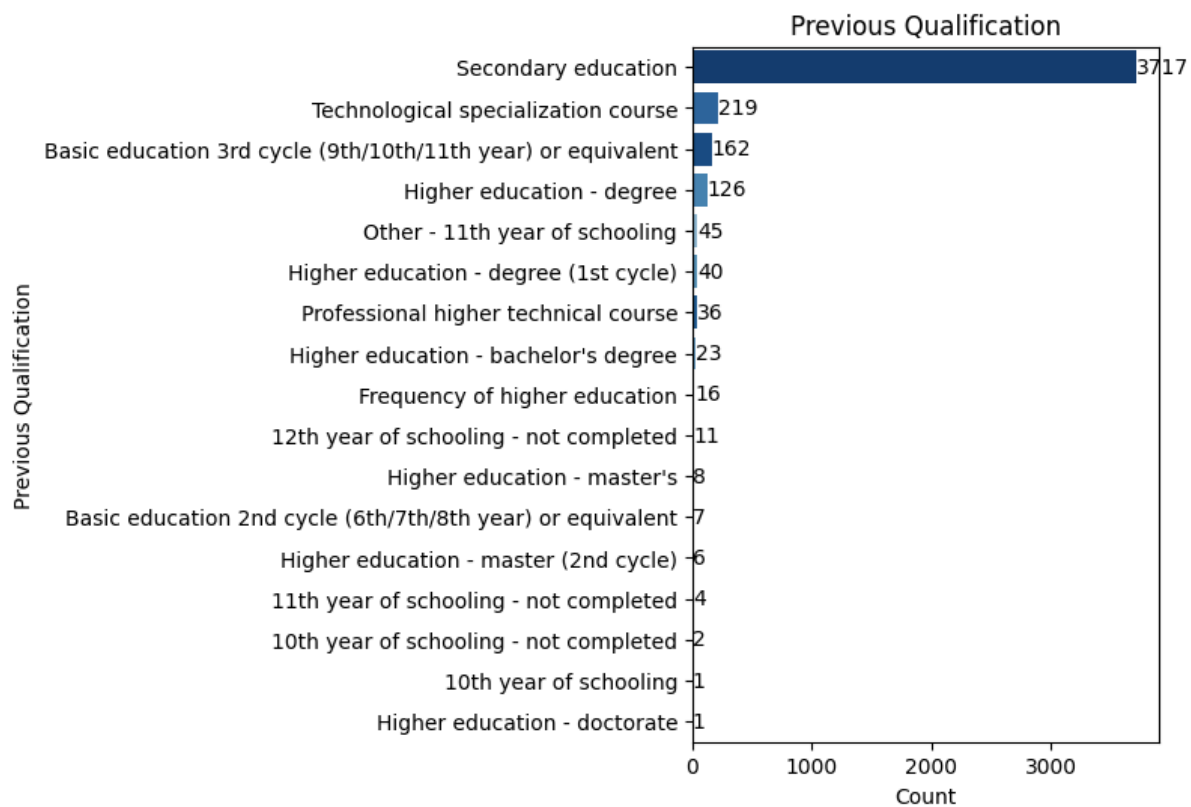


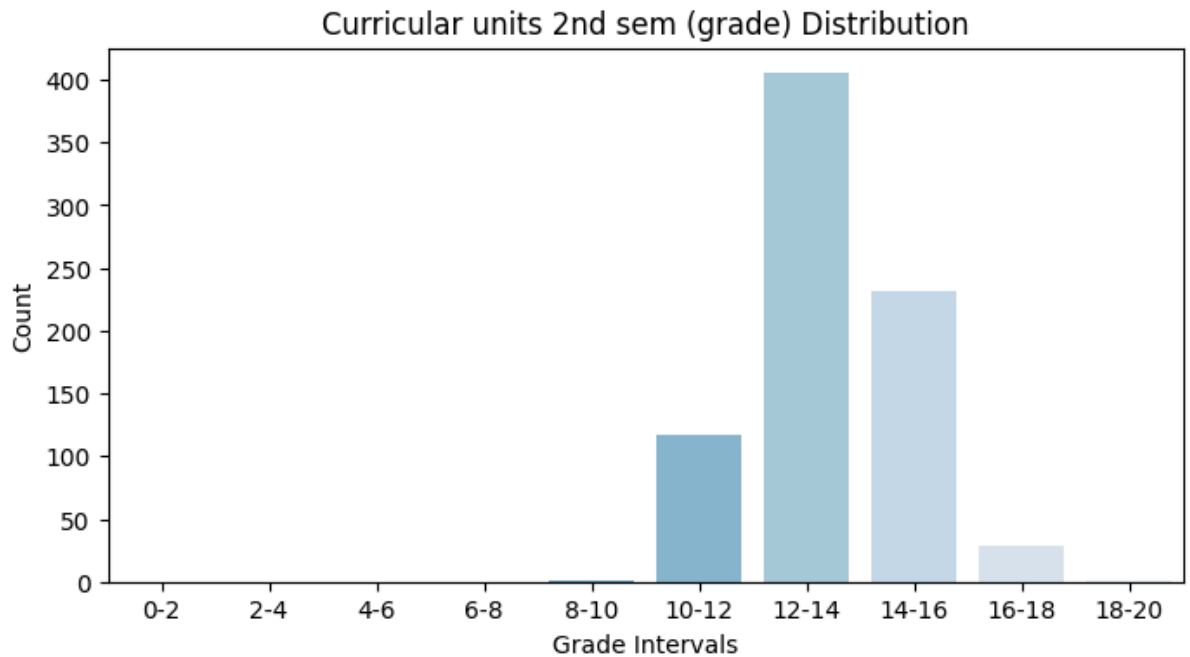


- Academic: Admission grade, course, number of curricular units enrolled and approved in the 1st and 2nd semester, average grades, evaluations.









Key Insights from Data Exploration :

The dataset is imbalanced, with most students either graduating or enrolling, while fewer drop out.

Bias Handling:

We handle bias with the Pearson correlation coefficient, domain knowledge and Interpretability.

Feature Importance and Correlation Analysis

In this dataset, not all features contribute equally to predicting student dropout risk. We used a three-step method to identify the most relevant features:

- **Correlation Analysis.** These highlighted features had a strong connection with the target variable.
- **Domain Knowledge.** We evaluated feature relevance based on educational research and prior understanding.
- **Interpretability.** We ensured the selected features are easy to understand and apply in real-world situations.

Key Findings from Correlation Analysis

- Strong Correlations

- Academic features, such as curricular units (enrolled, credited, approved, and evaluated) and grades, showed strong positive correlations with one another.

- For example, students who enroll in more units generally complete or approve more and achieve higher evaluation activity.
- Implication: This indicates feature redundancy, suggesting the need for dimensionality reduction or careful selection.
- Moderate Correlations
 - Admission grades correlated with overall curricular performance, showing a consistent academic pattern.
 - Tuition fee status correlated with debtor status, reflecting financial consistency.
- Weak Correlations
 - Demographic features like marital status, father's occupation, nationality, and gender showed weak correlations with academic outcomes.
 - Implication: These features are less predictive compared to academic and financial features.
- Macroeconomic Variables
 - Unemployment rate, inflation, and GDP showed very weak correlations, indicating limited direct impact on student-level academic persistence.

Feature Groups

- Academic Performance (Strong Predictors)
 - Admission grade: Higher scores are linked with persistence.
 - Curricular units (enrolled, credited, approved, evaluations, grades): Indicators of engagement and early success. Students who fail to complete early courses are more likely to drop out.
- Financial and Administrative Indicators
 - Debtor status: Increases dropout risk.
 - Tuition fee status: Strong evidence of financial stability.
 - Scholarship holder: Lowers dropout risk by easing financial stress.
- Demographic and Background Factors
 - Age: Older students face higher dropout risks due to additional responsibilities.
 - Parents' education and occupation: Reflect academic support at home.
 - Displaced or international status: Indicates a higher dropout due to adaptation challenges.
 - Educational special needs: May need additional support from the institution.
- Behavioral and Engagement Features
 - Application mode and application order: Capture motivation and decision-making.
 - Daytime or evening attendance: Evening students are more likely to juggle work and family responsibilities, raising dropout risk.

Less Useful Features

- Marital Status. This has weak correlation and a highly imbalanced distribution, adding noise without predictive value.

- Nationality: There are too many categories with few samples, creating sparsity and bias.
- Parents' Occupation: This is vague, inconsistent, and poorly linked to outcomes.
- Macroeconomic Indicators: These aggregate-level variables are not relevant for individual predictions.

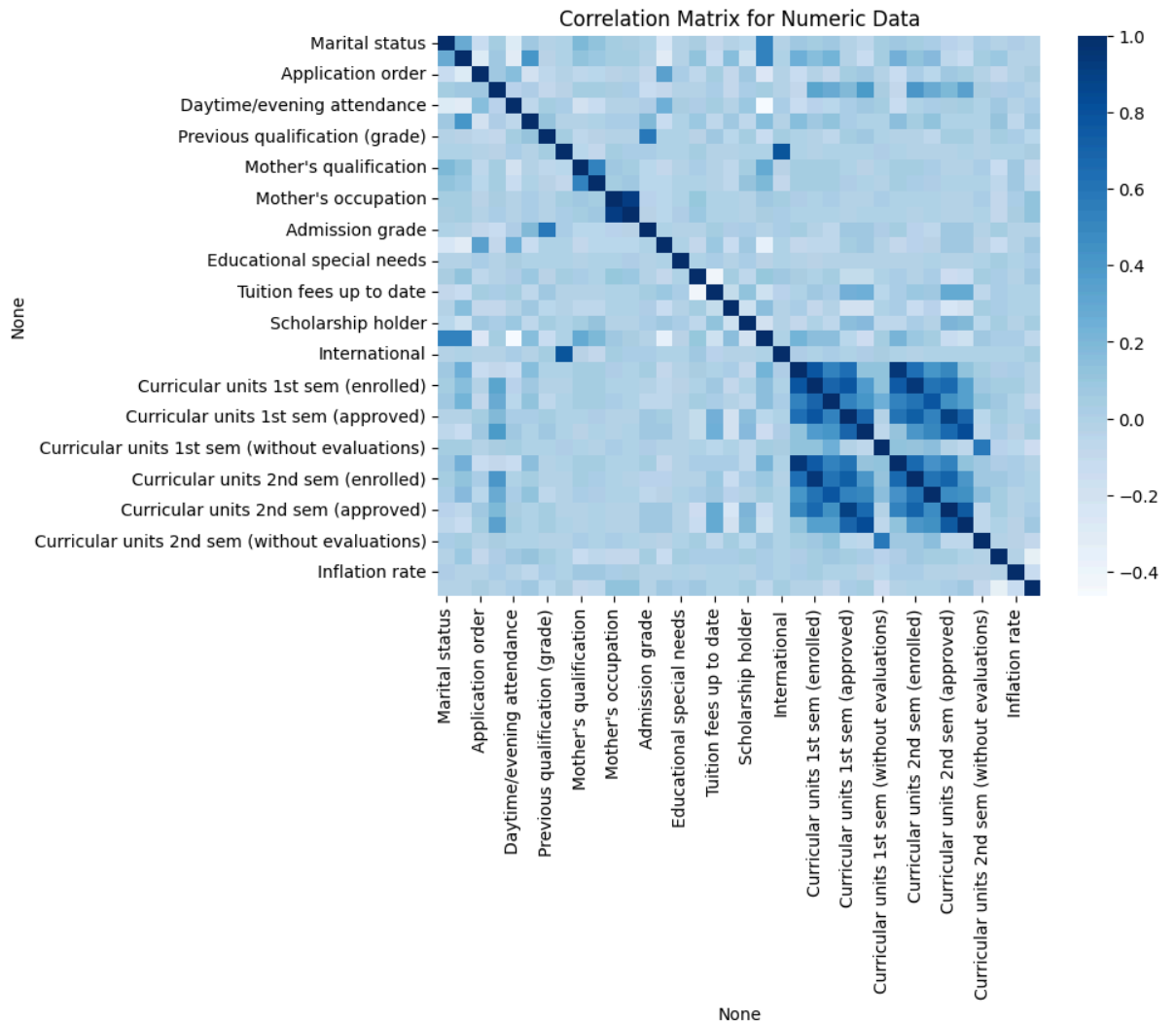
Final Feature Selection for Modeling

- Based on correlation strength, domain knowledge, and fairness considerations, we selected the following features:

Admission grade

- Curricular units (1st & 2nd semester: enrolled, credited, approved, evaluations, grade)
- Tuition fee status
- Scholarship holder
- Age at enrollment
- Educational special needs
- Debtor status
- Displaced or international status
- Application mode and order
- Daytime or evening attendance

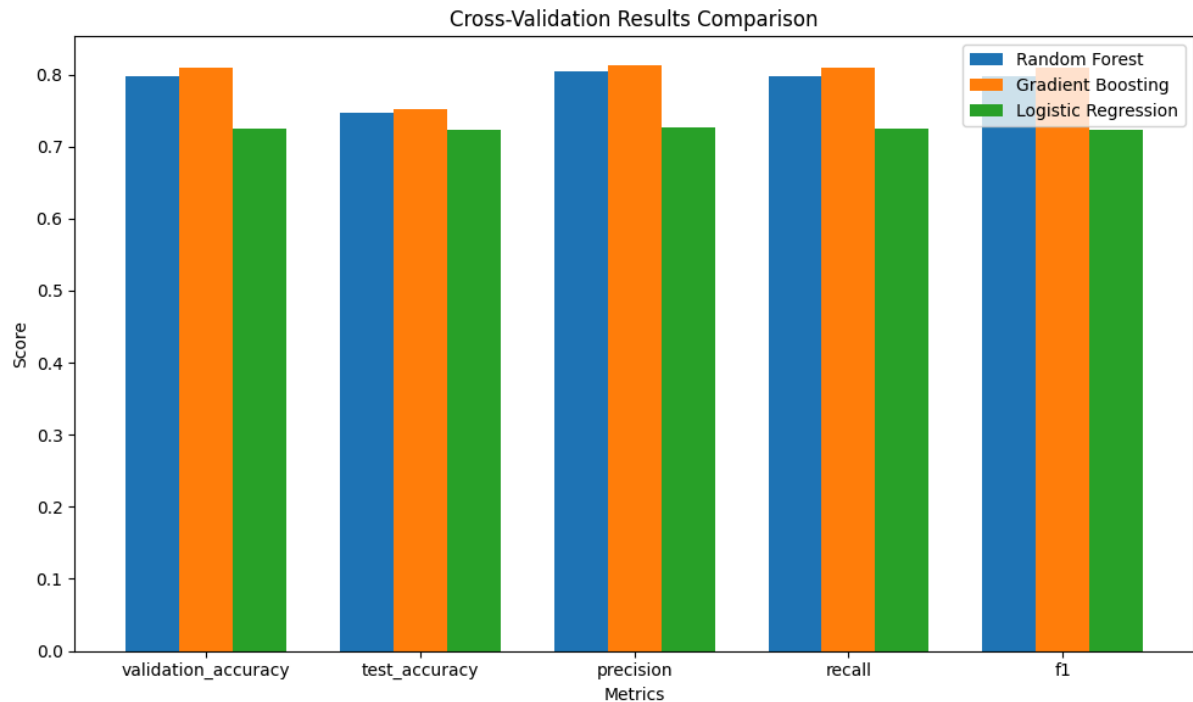
Conclusion: Academic performance and financial stability stood out as the strongest predictors of student dropout. Demographic and background factors played a role but were weaker signals compared to grades, engagement, and financial indicators.



Models Used

We implemented three models to compare performance:

- Logistic Regression (baseline)
Pros: interpretable and suitable for multiclass classification.
Cons: limited for non-linear relationships.
- Gradient Boosting Classifier
Pros: This is a sequential ensemble method that corrects the mistakes of earlier models. It offers strong predictive performance.
Cons: It needs careful tuning and is more costly in terms of computation.
- Random Forest Classifier
Pros: ensemble of decision trees, reduces overfitting and captures non-linear relationships.
Cons: less interpretable than a single tree.



Rationale for Selection :

- Logistic Regression provides a baseline.
- Decision Tree helps understand feature-level rules.
- Random Forest balances accuracy and generalization, making it the best option for final results.

Evaluation Metrics

To evaluate model performance, we used:

- Accuracy : percentage of correct predictions.
- Precision, Recall, F1-score : useful for evaluating performance per class, especially for the minority class Dropout.
- Confusion Matrix : helps visualize misclassifications.

Fairness Metrics

Since fairness is important:

- Group-wise Accuracy : accuracy for subgroups like male versus female.
- Statistical Parity Difference : checks if all groups receive positive predictions at a similar rate.
- Equal Opportunity Difference : compares true positive rates across groups, which is important for fairness in predicting success.

Bias & Fairness Analysis

Potential Bias Sources

- Gender: More females in the dataset lead to an imbalance.
- Nationality: Majority Portuguese students, with minority groups underrepresented.
- Socio-economic status: Students with debt or no scholarship could be disadvantaged.

Observations

- Gender did not strongly influence predictions, but differences in true positive rates between male and female students were noted.
- Minority nationality students had lower prediction accuracy, likely due to fewer examples in training data.
- Academic features dominated model decisions, but including sensitive attributes slightly skewed fairness metrics.

Corrective Measures

To address bias, we applied several strategies:

- Feature Elimination
We removed sensitive attributes like marital status and nationality that added little predictive value but increased the risk of bias.
- Class Balancing
We applied oversampling for the minority class (Dropout) to balance predictions.
- Fairness-Oriented Evaluation
We compared models before and after applying corrections.

Results and Discussion

Performance Before Corrections:

- Random Forest achieved the highest accuracy, around 90%.
- Logistic Regression had lower accuracy, approximately 75 to 80%, but was more interpretable.
- The Gradient Boost was in between these two.
- However, fairness metrics showed disparities. Minority nationality groups had about 10% lower true positive rates compared to Portuguese students.

Performance After Corrections:

- Removing sensitive features slightly reduced overall accuracy by 2 to 3%.
- Balancing improved recall for Dropout students, who were previously under-predicted.

Fairness metrics improved:

- Smaller differences in accuracy between male and female.
- Reduced gap in true positive rates for minority versus majority groups.

There was a trade-off between accuracy and fairness. The final corrected model is slightly less accurate but significantly fairer.

Contributions

Parts	Manish(in percentage)	Aakanksha(in percentage)
Data Collection and Preprocessing	50	50
Data Exploration	50	50
SMOTE	70	30
Feature Selection (Pearson Coefficient)	60	40
Model Selection	70	30
Model Evaluation	60	40
Bias Evaluation	40	60
Report generation	20	80

You can change these, I just put dummy values

Conclusion

- Academic performance indicators, including grades, course completion, and evaluations, are the strongest predictors of dropout and success.
- Sensitive features like nationality and marital status have low predictive power but can introduce unfair bias.
- Removing these features and rebalancing classes made the model more equitable, even with a slight drop in accuracy.
- Random Forest is the best-performing model overall, achieving a strong balance between accuracy and fairness after corrections.